

Weather Logger

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Presented here is a personal weather logger whose various transducers are used to log weather data such as humidity, atmospheric pressure and temperature. The data is transferred to PC where it is displayed as a graph for analysis.

The weather logger has the following features:

1. Measures and displays real-time value of humidity, pressure and temperature on the LCD
2. Saves data for 35 hours with sampling interval of 30 minutes in the microcontroller
3. Shows clock/calendar function
4. Has USB connectivity for data

PARTS LIST

Semiconductors:

- IC1 - PIC18F2550 microcontroller
- IC2 - DS1337 PC serial real-time clock
- IC3 - LM92 digital temperature sensor
- IC4 - SHT11 humidity sensor
- IC5 - MPX5100 pressure sensor
- IC6 - 7805, 5V regulator
- LED1 - 5mm LED
- LCD1 - 16×2 LCD
- BR1 - Bridge rectifier module, 1A
- D1-D4 - 1N4148 sw itching diode
- D5 - 1N4007 rectifier diode
- ZD1 - 5.1V, zener diode

Resistors (all 1/4-watt, ±5% carbon):

- R1 - 680-ohm
- R2, R10 - 4.7-kilo-ohm
- R3 - 100-ohm
- R4, R6-R9 - 10-kilo-ohm
- R5 - 1-mega-ohm
- VR1 - 10-kilo-ohm preset

Capacitors:

- C1 - 470µF, 25V electrolytic
- C2, C4, C9, C10, C12, C13 - 0.1µF ceramic
- C5 - 1µF, 16V electrolytic
- C6 - 470µF ceramic disk
- C7, C8 - 22pF ceramic disk
- C11 - 0.33µF ceramic disk

Miscellaneous:

- CONN1 - USB B-type connector
- SI-55 - Tactile switch
- X1 - 20MHz crystal oscillator
- X_{TAL2} - 32.768kHz crystal oscillator
- X_{TAL1} - 230V AC primary to 9V, 500mA secondary transformer
- BATT.1 - 9V battery



transfer to PC

5. Stores data in a PC for future analysis

6. Displays a graph in the PC as shown in Fig. 1

7. Sets clock and reference pressure using switches

Circuit and working

The circuit diagram of a weather logger is shown in Fig. 2. The circuit is built around microcontroller PIC18F2550 (IC1), serial real-time clock DS1337 (IC2), temperature sensor LM92 (IC3), humidity sensor SHT11 (IC4), pressure sensor MPX5100 (IC5), 5V voltage regulator 7805 (IC6) and 16×2 LCD display (LCD1).

The circuit requires 5V power

three external interrupts, four timer modules (Timer0-Timer3) and 10-bit ADC module with up to 13-channel multiplexer.

Real-time clock. The DS1337 serial real-time clock is a low-power clock/calendar chip with two programmable time-of-day alarms and a programmable square-wave output. Address and data are transferred serially through an I²C bus. The clock/calendar provides seconds, minutes, hours, day, date, month and year information. The date at the end of the month is automatically adjusted for months with fewer than 31 days, including corrections for leap year. The clock operates in either the 24-hour or 12-hour format with AM/PM indication.

Pins 3, 5 and 6 of serial real-time clock IC2 are connected to pins RB2, RB0 and RB1 of IC1, respectively, with pull-up resistors. At pins 1 and 2 of IC2, external crystal oscillator of

frequency 32.768 kHz is connected. External 9V battery is used to power the RTC.

Temperature sensor. LM92 is a digital temperature sensor and thermal window comparator with an I²C serial bus interface and an accuracy of ±0.33°C. The

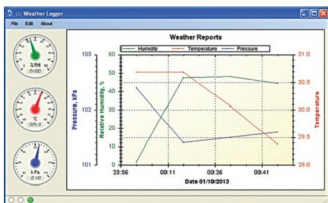


Fig. 1: Screenshot of weather logger graph

supply. The mains power supply is stepped down to 9V AC, 500mA by transformer X1. This stepped-down AC voltage is rectified by bridge rectifier BR1 and filtered by capacitor C1 before it is fed to IC6. Regulator IC6 provides regulated 5V DC supply. LED1 indicates the presence of power in the circuit.

Microcontroller. The PIC18F2550 microcontroller is the heart of this project. It works with a 20MHz crystal and has 32 kb of flash memory, 2 kb of RAM and 256 bytes of EEPROM. The controller is USB 2.0 compatible with speed of 1.5 Mbps to 12 Mbps. It has

window-comparator architecture of the LM92 eases the design of temperature control systems. Its open-drain interrupt (INT) output becomes active whenever temperature goes outside a programmable window, whilst a separate critical temperature alarm (T_{CRIT_A}) output becomes active when the temperature exceeds a programmable critical limit. The INT output can operate in either a comparator or event mode, whilst the T_{CRIT_A} output operates in the comparator mode only.

The host can program both the upper and lower limits of the window as well as the critical temperature limit.